SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA15 – CLOUD COMPUTING AND BIG DATA ANALYTICS

ASSIGNMENT 1 – CO1

Design, Analyse and Validate a Cloud-Based Solution for Migrating a Legacy University Management System

CO1: Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.

Field	Details	Field	Details
Student Name	V. Reddy Sreya	Register Number	192525124
Department / Section	BTech AIML	Academic Year / Semester	2025
Faculty Name	Dr T Sangeetha	Nature of Assignment	Individual
Duration	1 Week	Total Marks	50
Submission Date	02/11/2026	Evaluation Date	04/11/2026

MARK SHEET (Rubric-based evaluation)

Task	Title	Max	Awarded	Evidence / evaluation focus
1	Introduce Cloud Computing	3		
2	Trace Evolution and Hardware Evolution	3		
3	Analyse Internet Software Evolution	3		
4	Apply Server Virtualization	5		
5	Apply Web Services	4		
6	Analyse IaaS, PaaS and SaaS	5		
7	Examine XaaS	3		
8	Design Cloud Solution Architecture	5		
9	Hands-on Implementation and Documentation	4		
10	Evaluate and Recommend Final Solution	7		
11	Day-4 Design Challenge	8		
	TOTAL	**50**		**Percentage :**

EVIDENCE VERIFICATION CHECKLIST

Tasks	**Evidence expected**	**Status (✓ / P / ✗)**	**Evaluator note**
1–3	Conceptual explanations, comparisons and relevant illustrations		
4	Virtualization calculations with stated assumptions and peak-load caution		
5–7	API examples, service responsibility table and justified XaaS choices		
8	Labelled architecture diagram, caption below figure and component justification		
9	Actual code, environment details, execution logs, API outputs and screenshots		
10	Cost assumptions, 3-year totals, risk controls and migration roadmap		
11	Day-4 changes, 10× load strategy, 99.9% target and 40-college isolation		

IMPLEMENTATION EVIDENCE CHAIN

Evidence item	Status	**Observation**
Source code		
Virtualization		
PaaS or SaaS		
Functional test		
Failure case		
Screenshots		
Reproducibility		

OVERALL REMARKS

_____ **Student Signature**	_____ **Faculty Signature**	_____ **Program Director**

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = Present in documented answer; P = Partly present; ✗ = Missing / not satisfied

Task	Evidence expected	Status	Evaluator note
1	Cloud definition, characteristics, deployment models and scenario mapping		
2	Evolution timeline, hardware development and university relevance		
3	Internet software evolution, SOA and APIs		
4	Hypervisors, VM/container comparison and consolidation calculations		
5	SOAP/REST comparison and three university APIs		
6	IaaS/PaaS/SaaS responsibility layers and application mapping		
7	XaaS variants and two justified selections		
8	Labelled architecture, scaling, security and disaster recovery		

9	Working container, PaaS/SaaS deployment, actual tests and screenshots		
10	Three-year cost comparison, risks and migration roadmap		
11	Impact analysis, 10× load, 99.9% availability and tenant isolation		

IMPLEMENTATION EVIDENCE CHAIN

Evidence item	**Status**	**Observation**
Source code		
Virtualization		
PaaS or SaaS		
Functional test		
Failure case		
Screenshots		
Reproducibility		
Interpretation		
Code and report consistency		

SUBMISSION CHECKLIST (Not Separately Marked)

Requirement	**Status**
All 11 tasks answered	☐Yes ☐ Partly ☐ No
Cloud concepts and comparisons explained	☐ Yes ☐ Partly ☐ No
Virtualization calculations included	☐ Yes ☐ Partly ☐ No
API examples and service models explained	☐ Yes ☐ Partly ☐ No
Cloud architecture diagram included	☐ Yes ☐ Partly ☐ No
Implementation and testing evidence provided	☐ Yes ☐ Partly ☐ No
Three-year cost analysis and roadmap included	☐ Yes ☐ Partly ☐ No
Day-4 design challenge completed	☐ Yes ☐ Partly ☐ No
Figures and screenshots included	☐ Yes ☐ Partly ☐ No
Table and figure numbering correct	☐ Yes ☐ Partly ☐ No
Viva recommended	☐ Yes ☐ No

TASK 1 – Introduce Cloud Computing and Map It to the Scenario

1. Definition

Cloud computing provides computing resources such as servers, storage, databases and software through the internet without purchasing physical infrastructure.

2. Five Essential Characteristics

Characteristic	Explanation
On-demand self-service	Resources are available whenever required.
Broad network access	Services can be accessed through the internet.
Resource pooling	Resources are shared among multiple users.
Rapid elasticity	Resources increase or decrease based on demand.
Measured service	Users pay according to resource usage.

3. Deployment Models

1. Public Cloud: Services are provided by third-party providers.
2. Private Cloud: Infrastructure is dedicated to one organisation.
3. Hybrid Cloud: Combines public and private clouds.
4. Community Cloud: Shared by organisations with similar requirements.

Proposed Model: Hybrid Cloud

Hybrid cloud is suitable because sensitive student records can be protected in a private environment, while public cloud resources can support LMS, email and web applications.

5. Existing Problems and Cloud Solutions

Existing Problem	Cloud Solution
Low server utilisation (15%)	Resource pooling
Peak CPU usage above 95%	Rapid elasticity

Server procurement takes 6–8 weeks On-demand self-service

5. Benefits and Limitations

Benefits:

- Reduces infrastructure costs.
- Supports scalability and availability.
- Provides backup and disaster recovery.
- Enables access from all campuses.

Limitations:

- Requires internet connectivity.
- May create security and privacy risks.
- Depends on cloud providers.
- Can cause vendor lock-in.

Conclusion: Hybrid cloud is suitable for the university because it provides cost efficiency, security and scalability.

TASK 2 – Trace the Evolution and Hardware Evolution

1. Computing Evolution Timeline

Mainframe → Minicomputer → Personal Computer → Client–Server → Cluster Computing → Grid Computing → Utility Computing → Cloud Computing.

2. Evolution of Computing

Stage	Improvement
Mainframe	Centralised processing for multiple users.
Minicomputer	Reduced computing cost.
Personal Computer	Provided individual computing.
Client–Server	Enabled centralised data sharing.
Cluster Computing	Combined multiple computers for better performance.
Grid Computing	Shared resources across different locations.
Utility Computing	Introduced pay-per-use computing.
Cloud Computing	Provides on-demand internet-based resources.

3. Hardware Evolution

- Multi-core processors: Support parallel processing.
- Commodity servers: Provide affordable computing power.
- High-capacity storage: Stores large amounts of data.
- High-speed networking: Enables fast communication.
- Data centres: Provide large-scale computing infrastructure.

4. University's Current Stage

The university currently follows the traditional client–server computing model because it operates 12 dedicated physical servers for different applications.

Conclusion: Hardware improvements in processing, storage and networking enabled the development of scalable cloud computing.

TASK 3 – Analyse Internet Software Evolution

1. Evolution of Internet Software

Stage	Description
Static Web	Displays fixed web pages.
Dynamic Web	Generates content based on user requests.
Web 2.0	Supports interaction and collaboration.
SOA	Provides reusable software services.
Mashups	Combines services from different sources.

Cloud Applications Delivers software through the internet.

2. Role of Internet Technologies

- Browsers: Access web applications.
- HTTP/HTTPS: Transfers data between clients and servers.
- AJAX: Updates webpage content without full reloading.
- APIs: Allow different applications to communicate.

3. Software Comparison

Feature	Desktop	Client–Server	Web-Based
Installation	Required	Sometimes required	Usually not required
Access	Local system	Network	Internet
Maintenance	Individual	Centralised server and clients	Centralised
Scalability	Low	Medium	High

4. Impact on University Applications

The university can convert its monolithic applications into web-based services. Students can access results, fees, library services and learning materials through browsers and mobile applications using APIs.

Conclusion: Internet software evolution supports modern cloud applications through browsers, HTTP, AJAX and APIs.

TASK 4 – Apply Server Virtualization

1. Definition

Server virtualization divides one physical server into multiple virtual machines (VMs). A hypervisor manages the allocation of CPU, memory and storage to these VMs.

2. Types of Hypervisors

Type	Description	Example
Type-1	Runs directly on hardware.	VMware ESXi
Type-2	Runs on a host operating system.	VirtualBox

3. Virtualization Techniques

- Full Virtualization: Provides complete virtual hardware.
- Para-Virtualization: Guest OS communicates with the hypervisor.
- OS-Level Virtualization: Multiple containers share one OS kernel.
- Hardware-Assisted Virtualization: Uses processor features such as Intel VT-x and AMD-V.

4. Proposed Virtualization Plan

SIS	Virtual Machine
LMS	Containers
Library	Containers
Exam Portal	VM and Containers
Email	SaaS

VMs are suitable for legacy applications, while containers support lightweight and scalable applications.

5. Server Consolidation Calculation

Given:

- Physical servers = 12
- Average CPU utilisation = 15%
- Target host utilisation = 70%

Total CPU utilisation:

$$ 12\times15=180\% $$

Required hosts:

$$ \frac{180}{70}=2.57 $$
$$ \boxed{3\text{ physical hosts}} $$

Therefore, 12 physical servers can be consolidated into approximately 3 hosts, assuming equal server capacities and sufficient memory and storage.

6. Benefits and Risks

Benefits:

- Live migration moves VMs between hosts.
- Snapshots help restore previous VM states.
- Over-commitment improves resource utilisation.

Risks:

- Noisy neighbour affects performance.
- Hypervisor failure affects multiple VMs.
- Security isolation may be compromised.

Conclusion: Virtualization reduces hardware costs and improves resource utilisation by consolidating physical servers.

TASK 5 – Apply Web Services

1. Definition

Web services allow different applications to communicate and exchange information through the internet using standard protocols.

2. Web Service Technologies

Technology	Description
SOAP	Exchanges structured XML messages.
WSDL	Describes SOAP service operations.
UDDI	Helps discover web services.
REST	Uses HTTP methods and JSON/XML.

3. HTTP Methods

- GET – Retrieve information.
- POST – Submit information.
- PUT – Update information.
- DELETE – Remove information.

4. Proposed University Web Services

1. Student Result Lookup

Endpoint: /api/students/101/results

Method: GET

Sample response:

```
{
  "studentId": 101,
  "subject": "Cloud Computing",
  "grade": "A",
  "result": "Pass"
}
```

2. Fee Payment

Endpoint: /api/fees/payments

Method: POST

Sample request:

```
{
  "studentId": 101,
  "amount": 15000
}
```

3. Library Search

Endpoint: /api/library/books?name=Cloud

Method: GET

Sample response:

```
{
  "bookId": 201,
  "bookName": "Cloud Computing",
  "availability": "Available"
}
```

5. SOAP and REST Comparison

SOAP is suitable for legacy enterprise systems requiring formal message contracts. REST is lightweight and suitable for mobile applications and web services.

Conclusion: REST APIs are recommended for student results, fee payments and library searches because they are simple, flexible and easy to integrate.

TASK 6 – Analyse the Service Models: IaaS, PaaS and SaaS

1. Definition

- IaaS (Infrastructure as a Service): Provides virtual machines, storage and networking. Example: Amazon EC2.
- PaaS (Platform as a Service): Provides a platform to develop and deploy applications. Example: Google App Engine.
- SaaS (Software as a Service): Provides ready-to-use software through the internet. Example: Google Workspace.

2. Responsibility Comparison

Layer	**IaaS**	**PaaS**	**SaaS**
Hardware	Provider	Provider	Provider
Networking	Provider	Provider	Provider
Operating System	Customer	Provider	Provider
Runtime	Customer	Provider	Provider
Application	Customer	Customer	Provider
Data	Customer	Customer	Shared responsibility

3. University Application Categorisation

SIS	IaaS	Supports existing legacy applications.
LMS	PaaS	Supports application development and scaling.
Library System	PaaS	Provides easy database integration.
Exam Portal	IaaS/PaaS	Supports legacy systems and scalable services.
Email	SaaS	Reduces server maintenance.

4. Comparison of Service Models

Factor	**IaaS**	**PaaS**	**SaaS**
Control	High	Medium	Low
Cost	Usage-based	Usage-based	Subscription
Scalability	High	High	Provider-managed
Maintenance	High	Low	Very low
Vendor Lock-in	Moderate	Higher	Higher

Conclusion: The university should combine IaaS, PaaS and SaaS to support legacy applications, modern development and ready-to-use services.

TASK 7 – Examine XaaS

1. Definition

XaaS stands for Anything-as-a-Service. It provides different IT resources and services through the internet on demand.

2. Types of XaaS

Type	Description
DaaS	Provides virtual desktops.
DBaaS	Provides managed databases.
FaaS	Executes functions when required.
MBaaS	Provides backend services for mobile applications.
DRaaS	Provides disaster recovery services.

3. Two Recommended XaaS Services

1. Database as a Service (DBaaS):

It stores student records, examination results and fee details. It reduces database maintenance and provides automatic backups.

2. Disaster Recovery as a Service (DRaaS):

It maintains backups in another cloud region and helps restore university applications during failures.

4. Risks of XaaS

- Dependence on multiple providers.
- Data privacy and security issues.
- Increased service management complexity.
- Vendor lock-in and migration costs.

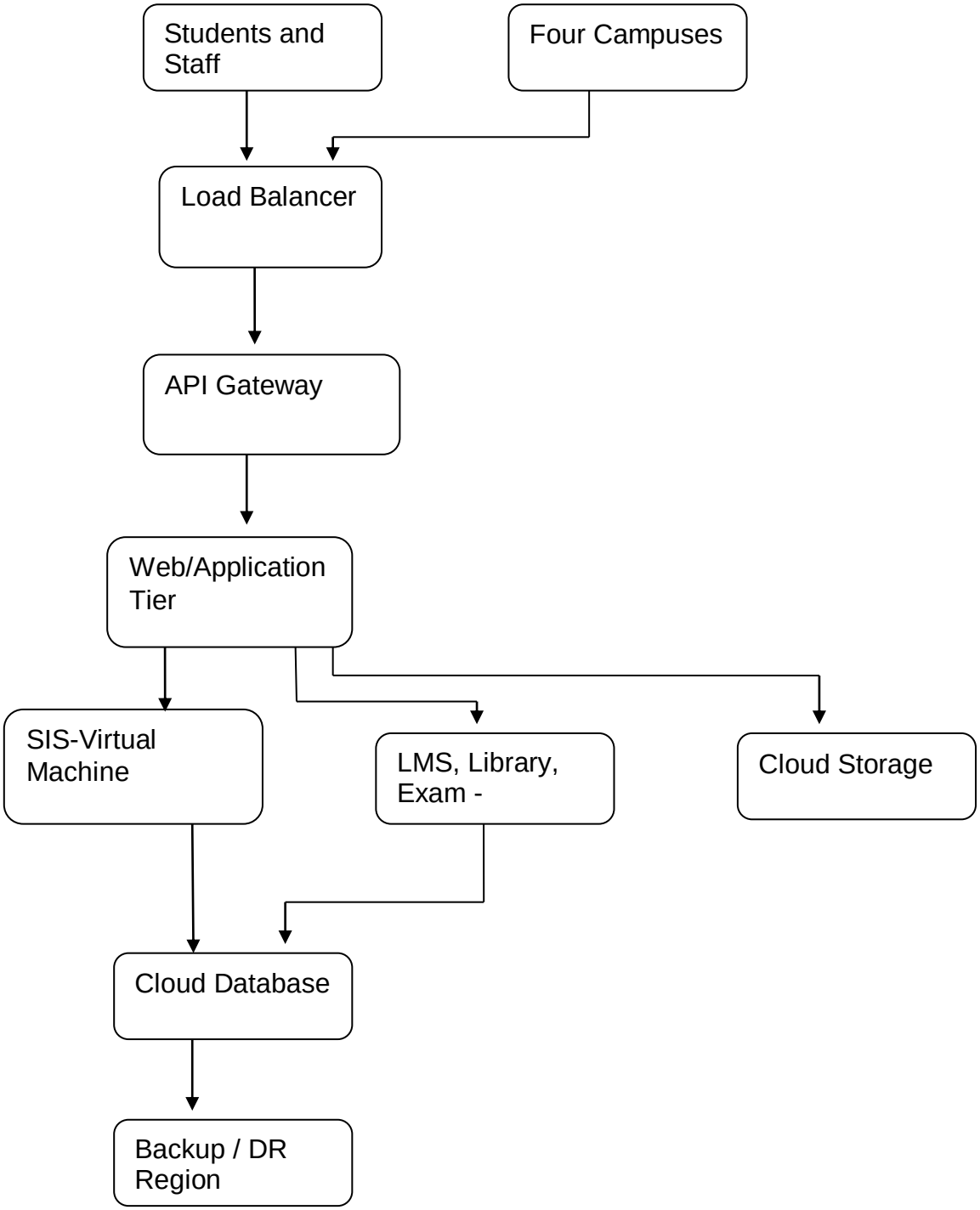
Conclusion: DBaaS and DRaaS improve database management, data protection and business continuity.

TASK 8 – Design the Cloud Solution Architecture (5 Marks)

1. Proposed Architecture

The university should adopt a hybrid cloud architecture combining private infrastructure with public cloud services.

2. Cloud Architecture Diagram



3. Components and Service Models

Component	Service Model	Purpose
Load Balancer	IaaS	Distributes traffic.
API Gateway	PaaS	Manages API requests.
Application Tier	IaaS/PaaS	Runs university applications.
Virtual Machines	IaaS	Hosts legacy systems.
Containers	PaaS	Runs modern applications.
Database	DBaaS	Stores university records.
Storage	Storage as a Service	Stores files and documents.
DR Region	DRaaS	Supports recovery.

4. Elasticity

Auto-scaling increases resources during admissions and examination periods and reduces them during normal periods. This improves performance and reduces unnecessary costs.

5. Security and Availability

- Use role-based access control.
- Encrypt data during storage and transmission.
- Maintain regular backups.
- Deploy applications across multiple availability zones.
- Monitor application performance and failures.

Conclusion: The proposed architecture provides scalability, security, availability and efficient resource management.

TASK 9 – Hands-on Implementation and Documentation (4 Marks)

1. Objective

To create a simple university student information REST API using Python, Flask and Docker.

2. Tools Required

Tool	Purpose
VS Code	Writing code
Python	Programming
Flask	Creating REST API
Docker	Containerisation
Browser	Testing API

3. Implementation

Step 1: Create a folder named UniversityCloudAPI.

Step 2: Create app.py and write:

```
from flask import Flask, jsonify

app = Flask(__name__)

@app.route('/students/101')

def student():

    return jsonify({

        "student_id": 101,

        "name": "Sreya",

        "department": "AIML",

        "status": "Active"

    })

app.run(host='0.0.0.0', port=5000)
```

Step 3: Create requirements.txt.

```
flask
```

Step 4: Create a Dockerfile.

```
FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]
```

Step 5: Build the Docker image.

```
docker build -t university-api .
```

Step 6: Run the container.

```
docker run -p 5000:5000 university-api
```

Step 7: Open the browser and enter:

```
http://localhost:5000/students/101
```

4. Expected Output

```
{
  "student_id": 101,
  "name": "Sreya",
  "department": "AIML",
  "status": "Active"
}
```

5. PaaS Deployment

The same application can be deployed using Render:

1. Upload the project to GitHub.
2. Create a Render account.
3. Select New Web Service.
4. Connect the GitHub repository.
5. Set the build command as `pip install -r requirements.txt`.
6. Set the start command as `python app.py`.
7. Deploy and test the generated URL.

6. Screenshots Required

- Python source code.
- Docker image creation.
- Running container.
- REST API output in browser.
- Cloud deployment output.

Conclusion: This implementation demonstrates a container-based REST API that can be deployed locally and hosted using a PaaS provider.

TASK 10 – Evaluate and Recommend the Final Solution

1. Comparison of Cloud Options

Factor	On-Premise	IaaS	PaaS/SaaS
Cost	High CapEx	Usage-based OpEx	Subscription / OpEx
Scalability	Low	High	High
Availability	Limited	High	Provider-managed
Security	Full physical control	Shared responsibility	Shared responsibility
Maintenance	High	Medium	Low

2. Three-Year Cost Estimation

Assumed estimated costs in Indian Rupees:

Factor	On-Premise	IaaS	PaaS/SaaS
Cost	High CapEx	Usage-based OpEx	Subscription / OpEx
Scalability	Low	High	High
Availability	Limited	High	Provider-managed
Security	Full physical control	Shared responsibility	Shared responsibility

Calculations:

On-Premise:

$$ 60+(4\times3)+(12\times3)=108 $$

IaaS:

$$ 8+(18\times3)+(4\times3)=74 $$

PaaS/SaaS:

$$ 10+(14\times3)=52 $$

PaaS/SaaS provides the lowest estimated cost because it reduces hardware maintenance and administration.

3. Risks and Mitigation

Risk	Mitigation
Vendor lock-in	Use standard APIs and portable data formats.
Data privacy	Apply encryption and access control.

Downtime	Use multiple availability zones.
Skills gap	Provide cloud training.
Cyberattacks	Use firewalls and MFA.

4. Final Recommendation

The university should adopt a hybrid cloud model using IaaS, PaaS, SaaS and XaaS because it provides better scalability, security, availability and cost efficiency.

5. Migration Roadmap

1. Assess existing applications.
2. Migrate Email to SaaS.
3. Migrate Library and LMS to PaaS.
4. Migrate SIS to IaaS.
5. Migrate Exam Portal to scalable cloud services.
6. Perform testing, backup and monitoring.

Conclusion: Hybrid cloud is the recommended solution because it reduces infrastructure costs and supports future university requirements.

TASK 11 – Re-evaluate and Adapt the Design

1. Impact Analysis

The university plans to launch a mobile application and provide student-record and fee-payment services to 40 affiliated colleges. The expected workload increases to 10 times the current peak, with 99.9% availability.

Changes in Requirements

Original Requirement	**Changed Requirement**
25,000 students and 2,000 staff	40 affiliated colleges added
Web applications	Web and mobile applications
Normal peak load	10 times current peak
Limited external access	Public APIs
No availability target	99.9% availability

Impact:

- Increased traffic requires additional computing resources.
- APIs become essential for mobile and college integration.
- Multi-tenancy requires separate access for each college.
- Higher availability requires backup and failover facilities.
- Increased workload may increase cloud costs.

2. Evaluation of Original Design

A. Service Models: IaaS, PaaS and SaaS can support the new requirements, but DBaaS and FaaS should also be introduced.

B. Virtualization: The original three-host plan may not support the new peak workload. Container orchestration and auto-scaling are required.

C. Availability Calculation:

Required availability = 99.9%.

8760x(100-99.9/100)

=8.60hours/year

A single-region deployment is insufficient protection against regional failures. Multi-zone deployment and secondary-region recovery are recommended.

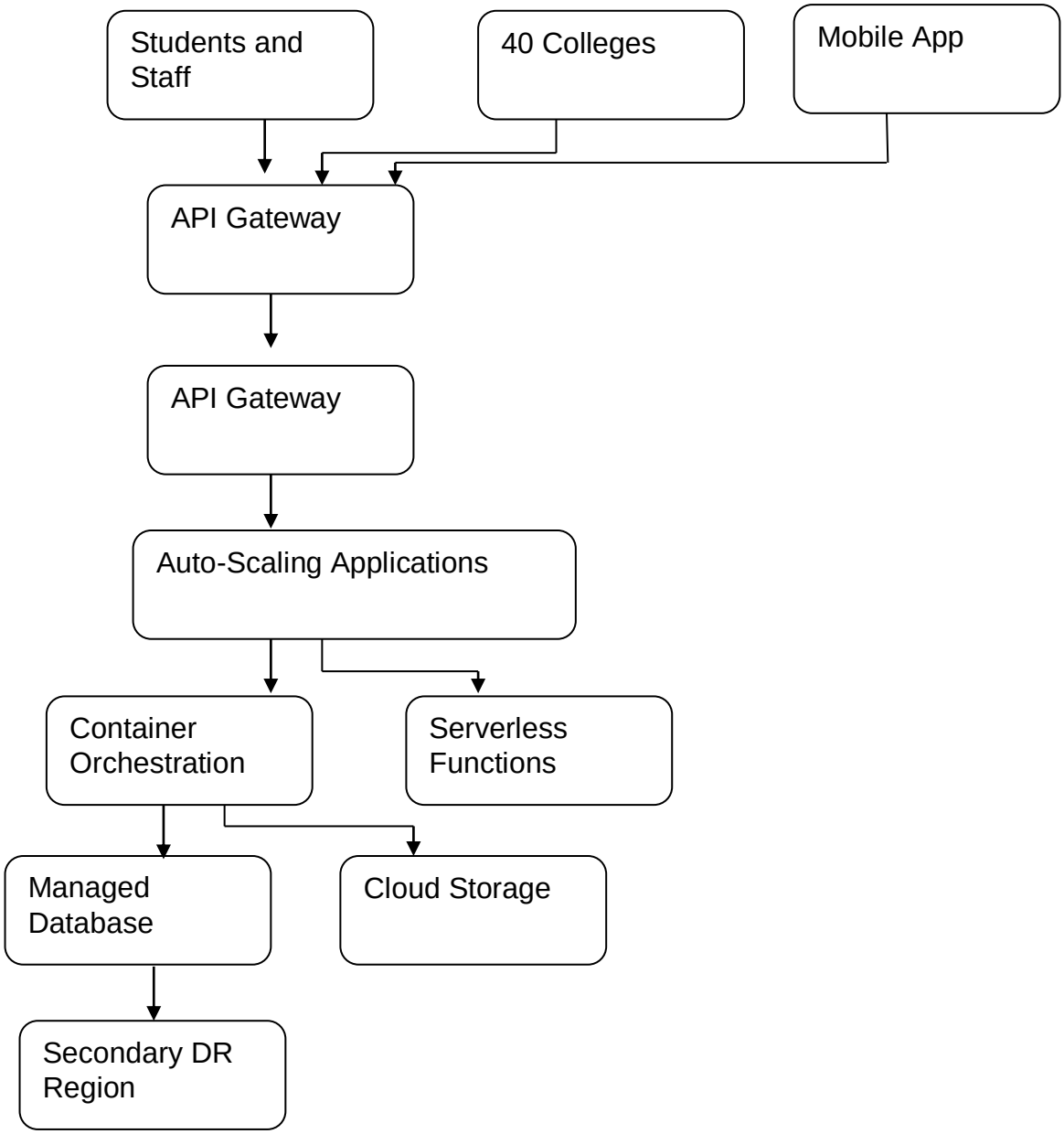
D. Cost: Increased traffic requires additional resources. Auto-scaling and serverless services can help control operating costs.

3. Adaptation of Original Design

Decision: MODIFY

The original hybrid cloud design should be modified to support higher traffic, mobile applications and affiliated colleges.

Revised Architecture



Major Changes:

1. API Gateway manages mobile and college requests.
2. Container orchestration supports automatic scaling.
3. Serverless functions handle notifications and background tasks.
4. Managed databases improve data management.
5. Multi-zone deployment improves availability.
6. Secondary-region backup supports disaster recovery.

4. Evidence-Based Defence

A. Capacity Calculation

Assume all 12 servers reach 95% CPU utilisation during the current peak.

Current peak CPU demand:

$$12x95\%=11.4$$

New demand at 10 times the peak:

$$114x10=114$$

Required host equivalents at 70% utilisation:

$$114/0.70=162.86$$

163 host equalents

This is a theoretical capacity estimate assuming equal server capacities. It does not mean purchasing 163 physical servers; cloud resources can provide equivalent virtual capacity.

B. Cost Comparison

Illustrative three-year estimates:

Cost Component	**Original Design**	**Revised Design**
Initial setup	₹8 lakh	₹12 lakh
Annual cloud cost	₹18 lakh	₹24 lakh
Annual backup and DR	₹4 lakh	₹6 lakh
Three-year total	₹74 lakh	₹102 lakh

Original:

$$ 8+(18\times3)+(4\times3)=74 $$

Revised:

$$ 12+(24\times3)+(6\times3)=102 $$

The revised design costs more but provides better scalability, availability and disaster recovery.

C. Security and Multi-Tenancy

- Give each college a unique organisation ID.
- Use role-based access control.
- Prevent colleges from accessing other colleges' records.

- Use encryption and multi-factor authentication.
- Secure APIs using authenticated tokens.
- Use a trusted payment gateway for fee transactions.

D. Trade-Offs

Advantages	**Trade-Offs**
High scalability	Increased cost
Better availability	More complexity
Secure API management	Additional configuration
Disaster recovery	Backup expenses
Container portability	Skilled administration required

Final Recommendation

The university should modify its original hybrid cloud architecture by introducing API Gateway, container orchestration, serverless computing, multi-zone deployment and secondary-region disaster recovery.

This solution supports the 10x workload, provides secure access to 40 affiliated colleges